

Dry Run.

Input

$n = 5$

flights = $[[0, 1, 200], [0, 2, 500], [1, 2, 100], [1, 3, 300],$
 $[2, 3, 100], [3, 4, 200]]$

src = 0, dst = 4, k = 2

Goal

Find cheapest price from $0 \rightarrow 4$ with at most 2 stops.

Step-by-Step Dry Run

• Step 1: Initialize

dist = $[0, \infty, \infty, \infty, \infty]$

• Iteration 1 (0 stops $\rightarrow$ direct neighbors)

Process all flights:

• $0 \rightarrow 1 \rightarrow \text{cost} = 200$

• $0 \rightarrow 2 \rightarrow \text{cost} = 500$

dist = $[0, 200, 500, \infty, \infty]$

• Iteration 2 (1 stop allowed)

From previous nodes:

• $1 \rightarrow 2 \rightarrow 200 + 100 = 300$ (better than 500 ok)

• $1 \rightarrow 3 \rightarrow 200 + 300 = 500$

• $2 \rightarrow 3 \rightarrow 500 + 100 = 600$ (ignore, worse)

dist = $[0, 200, 300, 500, \infty]$

• Iteration 3 (2 stops allowed)

• $2 \rightarrow 3 \rightarrow 300 + 100 = 400$ (better than 500 ok)

• $3 \rightarrow 4 \rightarrow 500 + 200 = 700$

dist = $[0, 200, 300, 400, 700]$

Final Answer

Cheapest cost = 700

Path Explanation

Best path: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

BUT stops = 3 $\rightarrow$ not allowed

So valid path: $0 \rightarrow 1 \rightarrow 3 \rightarrow 4 = 700$